

MH1200 Linear Algebra I.

Solutions to the Problem list for Week #2.

Core problems:

Problem 1:

Write down the augmented matrix for each of the following systems of linear equations.

$$(a) \quad \begin{array}{rcl} 7x_1 + x_2 + 4x_3 & = & -3 \\ 3x_1 - 2x_3 & = & 5 \\ -2x_2 + x_3 & = & 7 \end{array}$$

$$(b) \quad \begin{array}{rcl} 7x_1 + 2x_2 + x_3 - 3x_4 & = & 5 \\ x_1 + 2x_2 + 4x_3 & = & 1 \end{array}$$

$$(c) \quad \begin{array}{rcl} 2x_1 & = & 0 \\ 3x_1 - 4x_2 & = & 0 \\ x_2 & = & 1 \end{array}$$

$$(d) \quad \begin{array}{rcl} x_1 & = & 7 \\ x_2 & = & -2 \\ x_3 & = & 3 \\ x_4 & = & 4 \end{array}$$

Solution

$$(a) \quad \left(\begin{array}{ccc|c} 7 & 1 & 4 & -3 \\ 3 & 0 & -2 & 5 \\ 0 & -2 & 1 & 7 \end{array} \right)$$

$$(b) \quad \left(\begin{array}{cccc|c} 7 & 2 & 1 & -3 & 5 \\ 1 & 2 & 4 & 0 & 1 \end{array} \right)$$

$$(c) \quad \left(\begin{array}{cc|c} 2 & 0 & 0 \\ 3 & -4 & 0 \\ 0 & 1 & 1 \end{array} \right)$$

$$(d) \quad \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{array} \right)$$

Problem 2:

Which of the following augmented matrices are in row-echelon form.

$$(a) \quad \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 3 \end{array} \right)$$

$$(b) \quad \left(\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

$$(c) \quad \left(\begin{array}{cccccc|c} 1 & 0 & -2 & 0 & 2 & 0 & -2 \\ 0 & 1 & 0 & 0 & 2 & 0 & 4 \\ 0 & 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right)$$

Solution

- (a) This matrix is not in row echelon form because the leading entry of the third row is underneath the leading entry of the second row, whereas it should be strictly to its right.
- (b) This matrix is indeed in row echelon form because the leading entries move strictly to the right as you go down the rows.
- (c) This matrix is also in row echelon form.

Problem 3:

Consider the linear system:

$$\begin{aligned}x + y + 2z &= a \\x + z &= b \\2x + y + 3z &= c\end{aligned}$$

Show that for this system to be consistent, the constants a, b, c must satisfy $a + b = c$. (Hint: Recall that “consistent” means “has a solution”. Assume a solution exists and then try and show that the equation $a + b = c$ follows from these equations, which the solution satisfies.)

Solution: Assume that x, y and z are real numbers satisfying the equations above. If we add 1 copy of the first equation to 1 copy of the second equations and add (-1) times the third equation we deduce that

$$0x + 0y + 0z = a + b - c.$$

Thus assuming existence of a solution we deduce that $0 = a + b - c$.

□

Problem 4:

(From a 2020 Midterm exam.)

Consider the following system of linear equations in variables x_1, x_2, x_3, x_4 and x_5 :

$$\begin{aligned}2x_1 + x_2 - x_3 + x_5 &= 0 \\x_2 + x_4 + x_5 &= -1 \\2x_4 - 5x_5 &= 5\end{aligned}$$

Write down the augmented matrix for this system. Then use back-substitution to find a parameterization for the set of solutions of this system.

Solution.

The augmented matrix for this system is:

$$\left[\begin{array}{ccccc|c} 2 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 2 & -5 & 5 \end{array} \right].$$

According to the back-substitution procedure the variables x_3 and x_5 will become free parameters because the corresponding columns do not contain the leading entries of any rows. The variables x_1, x_2 and x_4 correspond to the leading entries of rows. We will use the corresponding rows to solve for these variables in terms of the free parameters.

We start by setting $x_3 = r$ and $x_5 = s$. Then, as is normal for back-substitution, we start at the last equation and work our way up.

The third equation says: $2x_4 - 5x_5 = 5$. We set $x_5 = s$ and this equation becomes:

$$2x_4 - 5s = 5 \Rightarrow x_4 = \frac{5}{2} + \frac{5}{2}s.$$

Now take the second equation. It says $x_2 + x_4 + x_5 = -1$. We solve for x_2 using the results for x_4 and x_5 we have already. We deduce:

$$x_2 = -1 - x_4 - x_5 = -1 - \left(\frac{5}{2} + \frac{5}{2}s\right) - s = -\frac{7}{2} - \frac{7}{2}s.$$

Finally the first equation says $2x_1 + x_2 - x_3 + x_5 = 0$. Solving this for x_1 using the results we have already we obtain:

$$x_1 = -\frac{1}{2}x_2 + \frac{1}{2}x_3 - \frac{1}{2}x_5 = -\frac{1}{2}\left(-\frac{7}{2} - \frac{7}{2}s\right) + \frac{1}{2}r - \frac{1}{2}s = \frac{7}{4} + \frac{5}{4}s + \frac{1}{2}r.$$

In summary, we have found a parameterization:

$$\begin{cases} x_1 = \frac{7}{4} + \frac{5}{4}s + \frac{1}{2}r \\ x_2 = -\frac{7}{2} - \frac{7}{2}s \\ x_3 = r \\ x_4 = \frac{5}{2} + \frac{5}{2}s \\ x_5 = s \end{cases} \quad r, s \in \mathbb{R}.$$

□

Problem 5:

Solve the following linear system by Gaussian elimination and back-substitution.

$$\begin{aligned}x_1 + x_2 + 2x_3 &= 8 \\-x_1 - 2x_2 + 3x_3 &= 1 \\3x_1 - 7x_2 + 4x_3 &= 10\end{aligned}$$

Solution: Start with the augmented matrix of the linear system and reduce it to row-echelon form by Gaussian Elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array} \right) \xrightarrow{\substack{R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 - 10R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & 0 & -52 & -104 \end{array} \right)$$

The linear system corresponding to this row-echelon form of the augmented matrix is:

$$\begin{aligned}x_1 + x_2 + 2x_3 &= 8 \\-x_2 + 5x_3 &= 9 \\-52x_3 &= -104\end{aligned}$$

Solving this by back-substitution gives:

$$\begin{aligned}x_3 &= \frac{-104}{-52} = 2, \\x_2 &= -9 + 5x_3 = -9 + 5(2) = 1, \\x_1 &= 8 - x_2 - 2x_3 = 8 - 1 - 2(2) = 3\end{aligned}$$

□

Problem 6:

Solve the following linear system by Gaussian elimination and back-substitution.

$$2x_1 - 3x_2 = -2$$

$$2x_1 + x_2 = 1$$

$$3x_1 + 2x_2 = 1$$

Solution

$$\left(\begin{array}{cc|c} 2 & -3 & -2 \\ 2 & 1 & 1 \\ 3 & 2 & 1 \end{array} \right) \xrightarrow{\begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - (3/2)R_1 \end{array}} \left(\begin{array}{cc|c} 2 & -3 & -2 \\ 0 & 4 & 3 \\ 0 & 13/2 & 4 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 - (13/8)R_2} \left(\begin{array}{cc|c} 2 & -3 & -2 \\ 0 & 4 & 3 \\ 0 & 0 & -7/8 \end{array} \right)$$

From the form of the row-echelon matrix we see that there are no solutions, i.e. the linear system is inconsistent. (To see this, note that the 3rd row of the row-echelon matrix corresponds to the equation $0 = -7/8$.)

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Problem 7:

Consider the following system of linear equations in three variables x_1, x_2 and x_3 :

$$x_1 + 0x_2 - 5x_3 = 1$$

$$0x_1 + 10x_2 + 2x_3 = 7$$

- (i) Write down the augmented matrix for this system.
- (ii) Is this matrix in row echelon form?
- (iii) Give a parametrization for the set of solutions of this equation.
- (iv) Briefly describe a geometric interpretation for this set of solutions. (But there is no need to draw a diagram.)

Solution

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & 1 \\ 0 & 10 & 2 & 7 \end{array} \right]$$

Yes, this is in row echelon form because the two rules for row echelon form are obviously satisfied.

Because it is in row echelon form we can use back-substitution to immediately write down a parametrization for the complete set of solutions. Column three doesn't contain a leading entry, so we set the corresponding variable to a free parameter $r \in \mathbb{R}$.

$$x_3 = r.$$

Then we solve the two equations to deduce:

$$x_2 = \frac{7}{10} - \frac{2}{10}r$$

and

$$x_1 = 1 + 5r.$$

The complete set of solutions is thus:

$$\left\{ \left(1 + 5r, \frac{7}{10} - \frac{2}{10}r, r \right), r \in \mathbb{R} \right\}.$$

If we rewrite this expression we get:

$$\left(1 + 5r, \frac{7}{10} - \frac{2}{10}r, r \right) = \left(1, \frac{7}{10}, 0 \right) + r \left(5, -\frac{2}{10}, 1 \right).$$

A geometric interpretation is: this is a straight line in $\mathbb{R}^3$ going through the point $(1, \frac{7}{10}, 0)$ in the direction $(5, -\frac{2}{10}, 1)$.

□

Problem 8:

A quadratic function is a function of the form

$$y = ax^2 + bx + c$$

where $a \neq 0$, b , and c are real constants.

There are three parts below. In each part, determine how many quadratic functions are there which have graphs going through all of the points (x, y) on the given list. Justify.

(i) $(-1, -2), (0, 1)$.

(ii) $(-1, -2), (0, 1), (1, 2)$.

(iii) $(-1, -2), (0, 1), (1, 2), (2, 1)$.

Solution

We let an arbitrary quadratic function be denoted by

$$y = ax^2 + bx + c.$$

There are many approaches to this. One is to first solve part (i). Another approach is to solve all these parts simultaneously. We'll do the parts one-by-one here.

(Part (i)): Each point contributes an equation. We obtain 2 equations:

$$\begin{array}{rcl} (-1, -2) & a - b + c & = -2. \\ (0, 1) & c & = 1. \end{array}$$

We recognise that this system is in row echelon form so we can solve it immediately. The general solution is: $c = 1$, $b = r$ a free parameter, $a = r - 3$. So for every choice of parameter $r \in \mathbb{R}$ we get a quadratic function going through the two points:

$$y = (r - 3)x^2 + rx + 1.$$

Because there is a free parameter, there are infinitely many different quadratic functions going through these two points, as we expect intuitively.

(**Part (ii)**): substitute the extra point $(1, 2)$ into this formula to get a condition on r . We get:

$$2 = (r - 3) + r + 1 = 2r - 2.$$

Thus $r = 2$. There is exactly one quadratic function going through these points, namely:

$$y = -x^2 + 2x + 1.$$

(**Part (iii)**): substitute the extra point $(2, 1)$ into this. We get:

$$1 = -4 + 4 + 1.$$

This equation is satisfied so the unique quadratic function going through the first three points also goes through this point.

In summary:

- (i) There are infinitely many quadratic functions.
- (ii) There is exactly one quadratic function.
- (iii) There is exactly one quadratic function.

□

Problem 9:

Let P be a plane in three dimensional space determined by the points $(1, 0, 1)$, $(1, 1, 0)$ and $(0, 1, 1)$. Find a linear equation in x, y, z with this plane as its solution set.

Solution

There are many ways to approach this problem. In this case it is even easy enough to guess the answer directly. The direct computational approach would be to introduce a general equation with new variables a, b, c, d representing the coefficients appearing in the equation.

$$ax + by + cz = d.$$

Every point now gives us an equation between the variables a, b, c , and d . The three equations we get are:

$$\begin{aligned} 1a + 0b + 1c - 1d &= 0 \\ 1a + 1b + 0c - 1d &= 0 \\ 0a + 1b + 1c - 1d &= 0 \end{aligned}$$

Now let's solve for a, b, c, d . We are searching for a solution different to the obvious solution $a = b = c = d = 0$ which gives a linear equation containing the given points but which is the whole space, not a plane.

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 0 \\ 1 & 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 2 & -1 & 0 \end{array} \right]$$

Now we can solve this system by back-substitution. The back-substitution procedure leads to d being made a free variable. The problem only asks us for 1 example of a linear equation giving this plane so we might as well go ahead and set the free variable d to 2. (Anything non-zero would be fine here. I chose 2 to make the first calculation easier.)

Back-substitution now gives: $c = 1$. Then $b = c = 1$. Then $a = d - c = 2 - 1 = 1$.

Thus, according to our calculations the following should be the desired equation:

$$x + y + z = 2.$$

Try and think of some other approaches to solving this problem. Later on when you meet some geometrical ideas like the cross-product, there will be a nice way to find this equation directly by finding two vectors in the plane, taking their cross-product, then taking the set of points perpendicular to that.

□

Problem 10.

Give an example of a system of linear equations which simultaneously satisfies all of the following conditions:

- There are 2 variables.
- Four different equations.
- Every entry in the augmented matrix is non-zero.
- There is exactly 1 solution.

Solution

Of course there will be many correct solutions to this. An easy way to construct an example is to start with any system of two equations in two variables satisfying these conditions, like:

$$\begin{cases} x_1 + x_2 = 1 \\ x_1 + 2x_2 = 1 \end{cases}$$

And then add any two different consequences of these equations. Such as:

$$\begin{cases} x_1 + x_2 = 1 \\ x_1 + 2x_2 = 1 \\ 2x_1 + 3x_2 = 2 \\ 3x_1 + 5x_2 = 3 \end{cases}$$

□

Problem 11:

Consider the following vector equation in three variables a , b and c :

$$(a + b)(1, -1, 1, -1) + (b + c)(1, 2, 3, 4) = (0, -3, -2, -5).$$

Write down a system of linear equations in the variables a , b and c that has the same set of solutions as this vector equation. (There is no need to solve the system.)

Solution

We get one linear equation for each entry of the vectors, by equating that entry on the two sides of the equation. To do this we need to first combine the expression on the left-hand-side into a single vector.

$$\begin{aligned} & (a + b)(1, -1, 1, -1) + (b + c)(1, 2, 3, 4) \\ &= (a + b, -a - b, a + b, -a - b) + (b + c, 2b + 2c, 3b + 3c, 4b + 4c) \\ &= (a + 2b + c, -a + b + 2c, a + 4b + 3c, -a + 3b + 4c). \end{aligned}$$

Equating this to the right hand side entry-by-entry we get a system of 4 linear equations in 3 variables:

$$\begin{cases} a + 2b + c &= 0 \\ -a + b + 2c &= -3 \\ a + 4b + 3c &= -2 \\ -a + 3b + 4c &= -5 \end{cases}$$

□

More abstract or challenging problems:

Problem 12:

Let L be the line in 3-dimensional space given by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} t - 6 \\ t - 10 \\ -2t + 20 \end{pmatrix} \quad t \in \mathbb{R}$$

Find a linear system that has L as its solution set.

Solution

We are asked to find a linear system. We'll use our intuition to guess the system, and then we can check that it is the correct guess. To begin we'll need to start thinking geometrically. A single linear equation in three variables determines a plane. If we want to determine a line, then we probably need a system of two linear equations with the planes corresponding to the separate equations intersecting in the desired line.

So we need to find two linear equations in the variables x, y, z that the points of this line satisfy. The trick to do this is to use the given equations to eliminate t , in two different ways. The first way is to take the difference $x - y$:

$$x - y = (t - 6) - (t - 10) = 4.$$

Or we add twice equation 2 to equation 3:

$$2y + z = 2(t - 10) + (-2t + 20) = 0.$$

So there is our answer:

$$\begin{aligned} x - y &= 4 \\ 2y + z &= 0. \end{aligned}$$

The planes corresponding to these two equations are not parallel (why?) so they intersect in a line. The line of intersection must be the given line because it is contained in both planes.

This argument is geometric and intuitive, so we can double check by solving this system. The system is already in row echelon form. If we set z to a free variable s we solve to find:

$$(x, y, z) = \left(4 - \frac{1}{2}s, -\frac{1}{2}s, s \right).$$

To see this gives the same set as in the problem substitute $s = -2t + 20$.

Then the formula becomes:

$$(x, y, z) = \left(4 - \frac{1}{2}(-2t + 20), -\frac{1}{2}(-2t + 20), -2t + 20 \right) = (-6 + t, t - 10, -2t + 20)$$

as required.

□

Problem 13:

The sum of the cubes of the first n natural numbers

$$C(n) = 1^3 + 2^3 + \cdots + (n-1)^3 + n^3 = \sum_{i=1}^n i^3$$

is expressible as a degree four polynomial in n :

$$C(n) = a_1 \cdot n + a_2 \cdot n^2 + a_3 \cdot n^3 + a_4 \cdot n^4 .$$

Set up a system of four linear equations that $a_1, \dots, a_4$ must satisfy and solve for them. Bonus: prove that your answer actually is the formula for the sum of the first n cubes!

Solution

First we set up a system of linear equations. Note that $C(1) = 1, C(2) = 9, C(3) = 36, C(4) = 100$. If $C(n) = a_1 \cdot n + a_2 \cdot n^2 + a_3 \cdot n^3 + a_4 \cdot n^4$ then a_1, a_2, a_3, a_4 must satisfy the following equations (given directly in augmented matrix form):

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 8 & 16 & 9 \\ 3 & 9 & 27 & 81 & 36 \\ 4 & 16 & 64 & 256 & 100 \end{array} \right]$$

OK, we have the equations set up. It's time for some Gaussian elimination:

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 4R_1 \end{array} \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 6 & 24 & 78 & 33 \\ 0 & 12 & 60 & 252 & 96 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 - 3R_2 \\ R_4 \rightarrow R_4 - 6R_2 \end{array} \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 6 & 36 & 12 \\ 0 & 0 & 24 & 168 & 54 \end{array} \right]$$

Finally:

$$R_4 \rightarrow R_4 - 4R_3 \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 6 & 36 & 12 \\ 0 & 0 & 0 & 24 & 6 \end{array} \right]$$

Now we have reached upper triangular form. Every column has a leading entry so there will be a unique solution. We solve by back substitution:

$$a_4 = \frac{1}{4}, a_3 = \frac{1}{2}, a_2 = \frac{1}{4}, a_1 = 0 .$$

Thus we obtain $C(n) = n^2/4 + n^3/2 + n^4/4 = (n(n+1)/2)^2$. This is the square of the n^{th} Triangular number! (Google this if you are interested.)

That completes the calculation if you accept the fact that the sum is indeed given by a degree 4 polynomial. But how would we go the extra step, and prove that this formula is correct for all n ? Let $f(n) = (n(n+1)/2)^2$. So far we have seen that $f(0) = C(0), \dots, f(4) = C(4)$, but we have not proven that $f(n) = C(n)$ for all $n \geq 1$. How do we know that also $f(5) = C(5)$?

If we show that $f(5) - f(4) = 5^3$ then this will show that $f(5) = C(5)$ as well. Why? Because then $f(5) = 5^3 + f(4) = 5^3 + C(4) = C(5)$. Similarly if we show that $f(6) - f(5) = 6^3$ this will imply that $f(6) = C(6)$.

To show that the formula is correct in general, we show that $f(n) - f(n-1) = n^3$ for any value of $n \geq 2$. This shows that $f(n) = C(n)$ for all values of $n \geq 1$, because

1. $f(1) = C(1)$
2. The fact that $f(n) - f(n-1) = n^3$ allows us to conclude that if $f(n-1) = C(n-1)$ then also $f(n) = n^3 + f(n-1) = n^3 + C(n-1) = C(n)$.

This is called a proof by *induction*.

Now let's do the algebra.

$$\begin{aligned} f(n) - f(n-1) &= \left(\frac{n(n+1)}{2} \right)^2 - \left(\frac{(n-1)n}{2} \right)^2 \\ &= \frac{n^2}{4} ((n+1)^2 - (n-1)^2) \\ &= n^3 . \end{aligned}$$

□

Problem 14:

In lectures we defined the concept of “row equivalence” which relates two matrices with the same shape (that is, the same number of rows and columns). We declared that two matrices with the same shape are row equivalent if there is a sequence of elementary row operations which takes you from one matrix to the other matrix. Explain why this does indeed define an equivalence relation on the set of matrices.

Discussion

If you have never encountered equivalence relations before, please learn about them as a first step in understanding this problem. They will be studied in MH1300.

To show any relation $\sim$ is an equivalence relation you have to demonstrate three things about it: reflexivity, symmetry, and transitivity. Here we'll just briefly explain why these three properties hold true here.

Reflexivity: you need to show that for every matrix M then $M \sim M$. This is basically by definition. When we say “sequence” we will allow empty sequences. If you do a sequence of zero moves to a matrix M you get back M . (There are other ways of seeing this too: you could do any row swap twice and you will get back the same matrix. Or any elementary move 3 where the constant is chosen to be zero.)

Symmetry: you need to show that whenever you have two matrices M and N , such that $M \sim N$, then $N \sim M$ as well. Well, the meaning of $M \sim N$ is that there is a finite sequence of elementary row operations $\mathcal{O}_i$ that starts with M and ends up with N . For the purposes of discussion assume there are k such operations.

$$M \xrightarrow{\mathcal{O}_1} M_1 \xrightarrow{\mathcal{O}_2} M_2 \dots \xrightarrow{\mathcal{O}_{k-1}} M_{k-1} \xrightarrow{\mathcal{O}_k} N.$$

We just need to explain why there is a going to be a sequence of operations getting back from N to M

$$N \xrightarrow{\mathcal{O}'_k} M_{k-1} \dots \xrightarrow{\mathcal{O}'_2} M_2 \xrightarrow{\mathcal{O}'_1} M.$$

This is quite easy to see. We just need to understand why each step in this sequence can be turned around. Each step in this sequence is an elementary row operation. There are three cases:

- (i) If B is obtained A by the elementary row operation $R_i \rightarrow cR_i$, with $c \neq 0$, then to go backwards, note that A can be obtained back from B by the elementary row operation $R_i \rightarrow \frac{1}{c}R_i$.
- (ii) If B is obtained from A by the elementary row operation $R_i \leftrightarrow R_j$, then A is obtained back from B by the same operation.
- (iii) If B can be obtained from A by the elementary row operation $R_i \rightarrow R_i + cR_j$, then A can be obtained back from B by the elementary row operation $R_i \rightarrow R_i - cR_j$.

Transitivity. This is the easiest case. You need to explain why if N can be obtained from M by a finite sequence of k elementary row operations $\mathcal{O}_i$, and if P can be obtained N by a finite sequence of l elementary row operations $\mathcal{O}'_j$, then there is a finite sequence of elementary row operations that take M into P . This is trivial: just join the two sequences into a longer sequence:

$$M \xrightarrow{\mathcal{O}_1} M_1 \xrightarrow{\mathcal{O}_2} M_2 \dots \xrightarrow{\mathcal{O}_k} M_{k-1} \xrightarrow{\mathcal{O}_{k-1}} N \xrightarrow{\mathcal{O}'_1} N_1 \xrightarrow{\mathcal{O}'_2} N_2 \dots \xrightarrow{\mathcal{O}'_l} P.$$

□

Problem 15:

Prove that it is impossible for a system of linear equations to have more than one but finitely many solutions. Hint: show that whenever you have two different solutions of a linear system, it is always possible to combine them somehow to construct new solutions.

Solution

We'll briefly explain how to see this and leave the student to check the details. Consider some system of linear equations in n variables. Assume we have two different solutions to the system:

$$(a_1, \dots, a_n) \quad \text{and} \quad (b_1, \dots, b_n).$$

Then you can check that for any real number $c \in \mathbb{R}$, then the vector

$$c(a_1, \dots, a_n) + (1 - c)(b_1, \dots, b_n)$$

is **also** a solution to the system.

Furthermore, as long as these two vectors are different, then this collection of solutions are all different solutions (i.e. when $c_1 \neq c_2$), so there must be infinitely many.

□